

REVIEW

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Recent advances in universal chimeric antigen receptor T cell therapy

Ning Jiang^{1,†}, Zhaoyi Yang^{1,†}, Huilei Miao^{1,†}, Shujun Xing¹, Shuhang Wang^{1*} and Ning Li^{1*}

Abstract

While chimeric antigen receptor (CAR) T cell therapy is highly effective for hematological malignancies, its widespread use is limited by complex, patient-specific manufacturing. Universal CAR-T (UCAR-T) cells, derived from allogeneic donors, offer a potential "off-the-shelf" solution. However, their clinical translation hinges on overcoming two key immunological barriers: graft-versus-host disease (GvHD) and host-versus-graft rejection (HvGR), which compromise safety and therapeutic persistence. This review summarizes recent advances in UCAR-T cell engineering and clinical strategies designed to improve both safety and efficacy. We discuss gene-editing technologies—such as CRISPR/Cas9 and base editors—used to prevent GvHD by ablating the T cell receptor (TCR) and to evade HvGR by disrupting human leukocyte antigen (HLA) expression. We also explore the development of UCAR-T products from alternative cell sources with low intrinsic alloreactivity, such as $\gamma\delta$ T cells. Furthermore, we detail multifaceted approaches to augment UCAR-T cell function and persistence, from the perspectives of enhancing intrinsic functions, reshaping the tumor microenvironment (TME) and overcoming tumor heterogeneity. Finally, we analyze recent clinical trial outcomes, which show promising efficacy in hematological malignancies but highlight ongoing challenges in solid tumors. The continued integration of sophisticated cellular engineering with innovative clinical strategies—such as enhanced lymphodepletion, combination therapies, and alternative administration routes—will be essential to realize the full potential of UCAR-T as a widely accessible and potent cell therapy.

Keywords UCAR-T cells, GvHD, HvGR, Efficacy, Solid tumor, Clinical trend, Clinical strategy

Introduction

Advances in our understanding of tumor immunology have led to the development of novel immunotherapies offering the potential for durable clinical benefit. In recent years, CAR T cell therapy has shown remarkable efficacy in treating relapsed or refractory (r/r) hematological malignancies, such as B-cell acute lymphoblastic

leukemia, diffuse large B-cell lymphoma, follicular lymphoma, and multiple myeloma. Numerous clinical trials have demonstrated that CAR T cell therapy can induce deep and durable remissions[1–5]. Currently, seven autologous CAR T cell products have received marketing approval in the United States. All of these products utilize viral vectors to introduce a second-generation CAR construct, which comprises an antigen-binding domain, a hinge region, a transmembrane domain, a co-stimulatory domain (derived from CD28 or 4-1BB), and a T cell activation domain (derived from CD3 ζ)[6].

Despite these successes, the real-world adoption of CAR T cell therapy has not kept pace with the number of approved products, raising concerns about the health economics and manufacturing logistics of this treatment modality[7–9]. The substantial cost of CAR T cell products, combined with the lengthy post-infusion

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monitoring period, places a significant financial and logistical burden on both patients and healthcare systems[10]. Moreover, the manufacturing lead time creates treatment delays that pose considerable risks, including disease progression and mortality. For instance, while awaiting B-cell maturation antigen (BCMA)-targeted CAR T cell therapy, 90% of patients experience disease progression, and 25% may succumb to their illness before the infusion can be administered[11]. These challenges are compounded by shortages of viral vectors and limited manufacturing capacity, which further restrict patient access[12]. Additionally, the production of autologous CAR T cells is susceptible to manufacturing failure or may yield a suboptimal product due to inter-patient variability, a compromised immune status, or an insufficient quantity of starting lymphocytes. The phenotype of the patient's T cells can also influence the therapeutic efficacy of the final product[13].

Even before the 2017 FDA approval of the first autologous CAR T cell therapy, research was underway to address the shortcomings of this approach by developing UCAR-T, also known as allogeneic or "off-the-shelf" CAR T cells. Derived from healthy allogeneic donors, these cells can be manufactured in large batches, cryopreserved, and administered to patients on demand[14]. This paradigm offers several key advantages: it significantly shortens the waiting period for patients, lowers manufacturing costs through economies of scale, and ensures consistent product quality. For patients in whom autologous manufacturing may not be feasible, such as those with T cell malignancies or individuals with poor T cell fitness from heavy pretreatment, UCAR-T cells provide a critical therapeutic alternative. Moreover, the "off-the-shelf" availability facilitates redosing for patients with an inadequate initial response, potentially improving overall treatment efficacy[15].

However, the use of allogeneic cells introduces unique safety and efficacy challenges. Donor T cells can attack host tissues, leading to a potentially fatal condition known as GvHD. Conversely, the host immune system can reject the UCAR-T cells in a HvGR, which limits their persistence and therapeutic effectiveness. Despite some advances, the clinical efficacy of UCAR-T cells, particularly concerning their persistence and the depth of response, often remains inferior to that of their autologous counterparts. Furthermore, when applied to solid tumors, UCAR-T cells confront the same formidable barriers as autologous CAR T cells, including inefficient trafficking to and infiltration of the tumor site, and poor function within the immunosuppressive TME.

This review provides an overview of emerging engineering strategies designed to improve the safety and efficacy of UCAR-T therapy. It will cover key approaches

for mitigating GvHD and HvGR, the use of alternative cell sources, and advanced modifications intended to enhance cell survival and function, especially, in solid tumors. Furthermore, this review will discuss the implementation of safety switches, analyze recent clinical trial trends, and examine the findings from representative studies.

Strategies to overcome GvHD, HvGR and fratricide

A central challenge in UCAR-T cell therapy is mitigating the bidirectional alloreactivity between the host and the graft. GvHD, where donor T cells attack recipient tissues, and HvGR, where the host immune system rejects the therapeutic cells, represent the primary hurdles to safety and persistence. To address these, a range of strategies involving both gene-editing and non-gene-editing techniques have been developed (Fig. 1; Table 1).

GvHD

The $\alpha\beta$ TCR expressed on allogeneic T cells is the primary mediator of GvHD; therefore, ablating the endogenous TCR is a cornerstone strategy for preventing this complication. Given the high sequence diversity within the variable region, gene editing strategies typically target the constant region. The T cell receptor alpha constant (TRAC) locus is a frequent target, as it is a single-copy gene with low homology to other genomic regions, making it more amenable to efficient editing. In contrast, the T cell receptor beta constant (TRBC) locus contains two highly homologous genes (TRBC1 and TRBC2). Targeting the TRAC locus is generally more efficient, as disrupting this single gene can effectively abolish surface TCR expression. For instance, zinc-finger nucleases (ZFNs)-mediated editing of TRAC resulted in TCR loss in 37% of T cells, compared to only 15% following TRBC editing[16]. Similarly, CRISPR/Cas9 editing achieved knock-out frequencies of 45% for TRAC and 15% for TRBC[48]. Conversely, targeting the TRBC locus presents several challenges. Incomplete knockout of both homologous TRBC genes can lead to residual TCR expression, thereby retaining the risk of GvHD. Furthermore, simultaneous editing of the two TRBC loci on chromosome 7 increases the potential for chromosomal rearrangements.

A more advanced manufacturing strategy involves the targeted insertion of the CAR transgene into a specific genomic locus. This approach overcomes the risk of insertional mutagenesis associated with the random integration of some viral vectors[49]. By directing the CAR construct to a specific site, such as the TRAC locus, this method ensures controlled CAR expression while simultaneously knocking out the endogenous gene responsible for TCR expression[50–52]. This can be achieved by combining gene-editing nucleases with a DNA repair

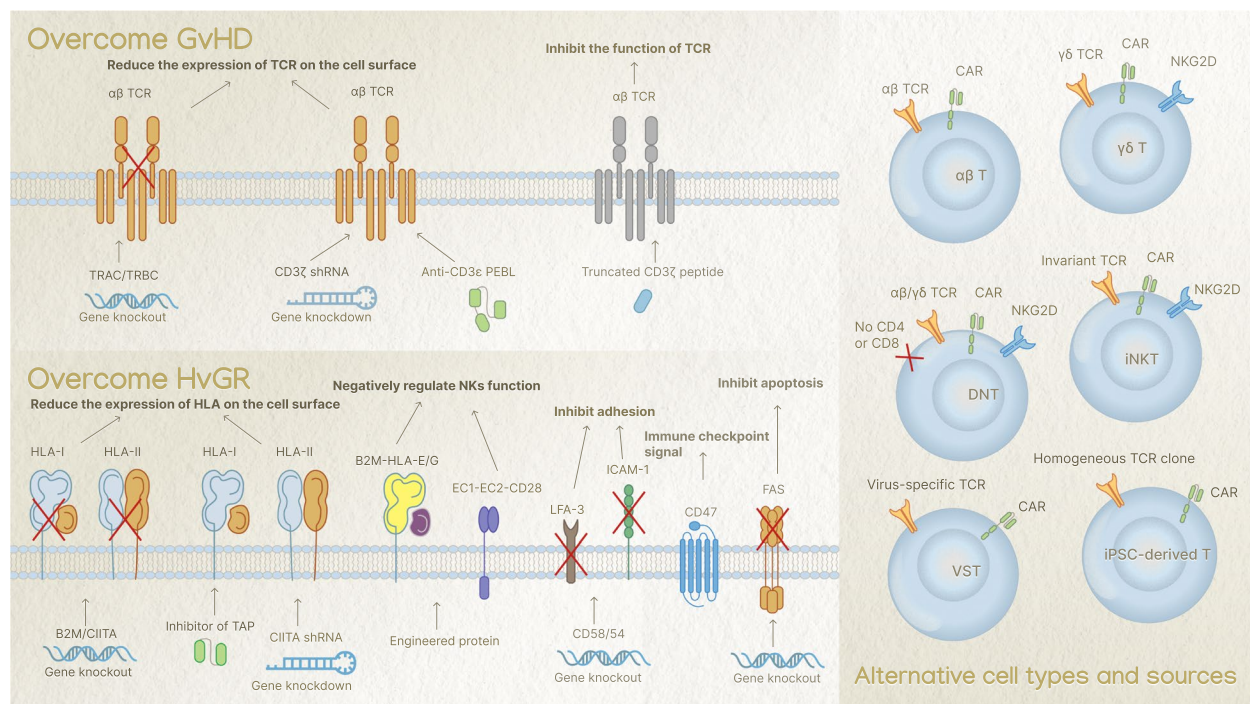


Fig. 1 Strategies to overcome GvHD and HvGR. To prevent graft-versus-host disease (GvHD), approaches focus on reducing endogenous T cell receptor (TCR) expression (e.g., TRAC/TRBC gene knockout, CD3 ζ shRNA gene knockdown, anti-CD3 ϵ PEBL) or inhibiting TCR function (e.g., truncated CD3 ζ peptides). To overcome host-versus-graft rejection (HvGR), strategies involve reducing Human Leukocyte Antigen (HLA) expression (e.g., B2M/CIITA gene knockout, CIITA shRNA gene knockdown, TAP inhibitors), negatively regulating Natural Killer (NK) cell function (e.g., engineered B2M-HLA-E/G, EC1-EC2-CD28), inhibiting adhesion (e.g., CD58/54 gene knockout), and preventing apoptosis (e.g., FAS gene knockout, CD47 modulation). Alternative T cell types and sources for universal CAR-T applications are also depicted, including $\gamma\delta$ T cells, invariant natural killer T cells (iNKTs), double-negative T cells (DNTs), virus-specific T cells (VSTs) and induced pluripotent stem cells (iPSCs)-derived T cells. These cells possess intrinsic properties that can mitigate alloreactivity while mediating distinct anti-tumor mechanisms

template delivered by a recombinant adeno-associated virus 6 (rAAV6) vector, facilitating precise integration at the desired genomic site[17, 24].

Following gene editing, residual $\alpha\beta$ TCR-positive T cells are commonly depleted using magnetic-activated cell sorting (MACS), a technique integrated into many manufacturing protocols. Although this enrichment step is not a manufacturing bottleneck, novel strategies are being explored to further enhance product purity. For instance, a recent study utilized a CAR natural killer (NK)-92 cell line targeting CD3 to eliminate remaining TCR-positive cells. This approach reportedly increased the total CAR T cell yield approximately threefold while preserving cytotoxic activity and a favorable cellular phenotype[23].

Beyond gene-editing, several non-gene-editing approaches are also being explored to prevent GvHD. One notable example is CYAD-211, a UCAR-T cell product engineered to co-express a short hairpin RNA (shRNA) targeting CD3 ζ , which reduces the surface expression of the TCR/CD3 complex[35]. Early clinical data have demonstrated a favorable safety profile and preliminary clinical activity, with no

evidence of GvHD[53]. In another study, researchers developed protein expression blockers (PEBLs) by fusing antibody-derived single-chain variable fragments (scFvs) specific for CD3 ϵ to distinct intracellular retention sequences. These PEBLs effectively eliminate surface TCR expression while preserving robust CAR T cell functions, including proliferation, cytokine secretion, and cytotoxicity against leukemia cells[36]. Furthermore, the UCAR-T cell product CYAD-101 employs a TCR inhibitory molecule (TIM) that competes with the endogenous CD3 ζ subunit, a strategy designed to reduce the risk of GvHD while maintaining the activity of its natural killer group 2 member D (NKG2D)-based CAR[37].

HvGR

HvGR is primarily driven by the recipient's immune system recognizing HLA class I (HLA-I) molecules on donor T cells. A common strategy to prevent this is to knock out the beta-2-microglobulin (B2M) gene, which encodes an essential component of the HLA-I complex. However, the resulting absence of HLA-I on UCAR-T cells triggers their destruction by host NK cells via a "missing-self" mechanism, mediated primarily by killer

Table 1 Strategies to overcome GvHD and HvGR

Refs	GvHD-preventing strategy	HvGR-preventing strategy	Gene-editing tool	Transduction	Target	Co-stimulatory domain
[16]	TRAC and TRBC knockout	NA	ZFN mRNA electroporation	SB transposon/transposase system	CD19	CD28
[17]	TRAC knockout (CAR)	B2M knockout (HLA-E)	TALEN mRNA electroporation	AAV	CD123/CD22	4-1BB
[18]	TRAC knockout	CD52 knockout; alemtuzumab	TALEN mRNA electroporation	Lentivirus	CD19	4-1BB
[19]	TRAC knockout	NA	TALEN mRNA electroporation	Lentivirus	CD123	4-1BB
[20]	TRAC knockout	CD52 knockout; alemtuzumab	TALEN mRNA electroporation	Lentivirus	CD19	4-1BB
[21]	TRAC knockout	CD52 knockout; anti-CD52 antibody	TALEN	Lentivirus	BCMA	4-1BB
[22]	TRAC knockout	CD52 knockout; anti-CD52 antibody	TALEN mRNA electroporation	Lentivirus	FLT3	4-1BB
[23]	TRAC knockout; TCR/CD3-CAR NK-92 cells	NA	Cas9/gRNA RNP electroporation	Lentivirus or dsDNA HDRT	CD19	CD28
[24]	TRAC knockout (CAR)	NA	Cas9 mRNA/gRNA electroporation	AAV	CD19	CD28
[25]	TRAC knockout	B2M knockout; CIITA knockout	Cas9/gRNA RNP electroporation	Lentivirus	CD19	4-1BB
[26]	TRAC knockout (CAR)	B2M knockout (HLA-E)	Cas12a/chRDNA RNP electroporation	AAV	BCMA	4-1BB
[27]	TRAC knockout	B2M knockout; mutant B2M-HLA-E expression; mutant B2M-HLA-G expression	Cas9/gRNA RNP electroporation	Lentivirus	CD19	4-1BB
[28]	TRAC knockout	HLA-A/B knockout	Cas9/gRNA RNP electroporation	Lentivirus	CD19	4-1BB
[29]	NA	B2M knockout; CD58 knockout; CD54 knockout	Cas9/gRNA RNP electroporation	Retrovirus	CD19	NA
[30]	TRAC knockout	B2M knockout; CIITA knockout	Cas9/gRNA RNP electroporation	Lentivirus	CD19	NA
[31]	TRAC knockout	B2M knockout; CIITA knockout	Cas9/gRNA RNP electroporation	Retrovirus	CD19	CD28
[32]	TRAC knockout	B2M knockout; CIITA knockout	Cas9/gRNA RNP electroporation	Lentivirus	Vβ2	4-1BB
[33]	TRAC knockout	RFX5 knockout; NKi expression	Cas9 mRNA/gRNA electroporation	Retrovirus	CD7	4-1BB
[34]	TRAC knockout	RFX5 knockout; two inhibitory receptors expression (i1, i2)	NA	NA	CD7	NA
[35]	TCR CD3ζ shRNA	NA	NA	Retrovirus	BCMA	4-1BB
[36]	anti-CD3ε PEBL	NA	NA	Retrovirus	CD19	4-1BB
[37]	TIM8	NA	NA	Retrovirus	NKG2D	NA
[38]	NA	TAPi; CIITA shRNA	NA	Lentivirus	CD19	NA
[39]	TCR knockout	SB9 overexpression	Cas9/gRNA RNP electroporation	Retrovirus	CD19	NA
[40]	TRAC knockout (mCNA and CAR)	CsA	Cas9/gRNA RNP electroporation	AAV	CD19	CD28
[41]	TRAC knockout	FAS knockout	Cas9/gRNA RNP electroporation	Lentivirus	CD19	4-1BB
[42]	TRBC knockout	B2M knockout	Cas-CLOVER mRNA/gRNA electroporation	piggyBac transposon	BCMA	NA
[43]	TRAC knockout	B2M knockout	Cas9/gRNA RNP electroporation	Lentivirus	CD19	4-1BB

Table 1 (continued)

Refs	GvHD-preventing strategy	HvGR-preventing strategy	Gene-editing tool	Transduction	Target	Co-stimulatory domain
[44]	TRAC base-editing knockout	B2M base-editing knockout	Base editor mRNA electroporation	Lentivirus	CD19	NA
[45]	TRAC knockout (CAR)	B2M base-editing knockout; CIITA base-editing knockout	Cas12a/TRAC gRNA electroporation; base editor mRNA electroporation	dsDNA HDRT	CD19	CD28
[46]	TRAC base-editing knockout	CD52 base-editing knockout; alemtuzumab	Base editor mRNA electroporation	Lentivirus	CD7	CD28
[47]	TRBC base-editing knockout	CD52 base-editing knockout; alemtuzumab	Base editor mRNA electroporation	Lentivirus	CD7	NA

Abbreviations: TRAC T cell receptor alpha constant, TRBC T cell receptor beta constant, NA not available, ZFN zinc finger nuclease, SB sleeping beauty, CAR chimeric antigen receptor, B2M beta-2-microglobulin, HLA human leukocyte antigen, TALEN transcription activator-like effector nuclease, AAV adeno-associated virus, BCMA B cell maturation antigen, FLT3 FMS-like tyrosine kinase 3, TCR T cell receptor, NK natural killer, Cas9 CRISPR-associated protein 9, RNP ribonucleoprotein, HDRT homology-directed repair template, CIITA class II transactivator, chdRNA CRISPR hybrid RNA-DNA, RFX5 regulatory factor X-5, NKI natural killer cell inhibitor, shRNA short hairpin RNA, PEBL protein expression blocker, TIM8 T cell receptor inhibitory molecule 8, NKG2D NK receptor group 2 member D, TAPi the transporter associated with antigen processing inhibitor, SB9 granzyme B-specific serine protease inhibitor SERPINB9, mCNA mutated calcineurin subunit A, CsA cyclosporine A, FAS factor-related apoptosis. Cas-CLOVER, a novel high-fidelity RNA-guided endonuclease

cell immunoglobulin-like receptors (KIRs)[54]. Several strategies have been developed to counteract this NK cell-mediated lysis. One approach involves engineering UCAR-T cells to express non-classical, inhibitory HLA molecules, such as HLA-E or HLA-G. Preclinical studies have validated this concept by inserting HLA-E into the B2M locus or by expressing B2M-HLA-E or B2M-HLA-G fusion proteins[17, 25–27]. An alternative method is the selective knockout of classical HLA-A and HLA-B genes while preserving HLA-E expression, which confers resistance to NK cell killing without impairing anti-tumor activity[28]. Other approaches include overexpressing multiple inhibitory receptors to resist lysis by NK cells and T cells[34], knocking out adhesion ligands like CD54 (ICAM-1) and CD58 (LFA-3) to improve persistence[29], and expressing CD47, a "don't eat me" signal, to protect against clearance by macrophages and NK cells[30, 55].

The role of HLA class II (HLA-II) molecules in HvGR is also a key consideration, as T cells can upregulate HLA-II expression upon activation[25, 31], rendering them susceptible to rejection by recipient CD4+ T cells[56]. Consequently, the class II major histocompatibility complex transactivator (CIITA), the master regulator of HLA-II transcription, has become a target for gene editing. For instance, UCAR-T cells with dual knockout of B2M and CIITA persisted for over a week in a murine T cell tumor model without exogenous cytokine support[32]. Another study found that HLA-II deficiency may enhance CAR T cell survival, an effect potentially linked to the upregulation of T cell proliferation pathways[25]. These findings suggest that the impact of HLA-II knockout on CAR T cell functionality warrants further investigation. More recently, a single-vector strategy was developed

to prevent rejection by co-expressing a viral inhibitor of the transporter associated with antigen processing (TAP) and an shRNA targeting CIITA, thereby simultaneously downregulating both HLA class I and class II expression[38].

Strategies targeting apoptosis pathways are also being explored as an alternative to HLA modification. Rejection by host T and NK cells is largely mediated by apoptosis, primarily through the granzyme B and Fas pathways. Accordingly, Teo et al. demonstrated that overexpressing an engineered SerpinB9 (SB9), a protease inhibitor targeting both granzyme B and caspases, could significantly reduce rejection[39]. More recently, a study showed that Fas knockout, in contrast to B2M knockout, rendered UCAR-T cells resistant to lysis by both T cells and NK cells. These Fas-deficient cells also exhibited superior control of leukemia growth in mice, suggesting Fas ablation as a promising alternative to B2M knockout[41]. Nevertheless, the safety of this strategy requires close scrutiny due to the critical role of Fas in maintaining immune homeostasis.

Another approach to mitigate HvGR is to combine UCAR-T cell therapy with pharmacological immunosuppression. This can be achieved by engineering UCAR-T cells for resistance to specific agents, such as cyclosporin A (CsA), allowing for their co-administration to suppress the host immune response[40]. Similarly, knocking out CD52 enables the use of potent lymphodepleting antibodies without eliminating the therapeutic cells, a strategy with extensive clinical precedent[21, 57, 58]. However, long-term pharmacologic immunosuppression poses considerable risks, including infection and drug-related toxicities. Consequently, such approaches may be most suitable when rapid, potent

immunosuppression is required for a short duration. For achieving durable, long-term UCAR-T cell persistence, gene-editing strategies like B2M knockout offer a more permanent solution. Nevertheless, the development of safer, long-term pre-infusion treatment regimens still remains an important goal for the field.

The combined knockout of TRAC and B2M is currently the most common genetic strategy in UCAR-T cell development (Table 1). However, a standard approach for preventing subsequent NK cell-mediated rejection has yet to be established. Nonetheless, several strategies are advancing clinically to address this, including the expression of an HLA-E fusion protein (CB-011, Caribou Biosciences) and the overexpression of inhibitory receptors (R13-02, Bioheng Biotech)[26]. A key challenge is that each additional gene edit increases manufacturing complexity and risk. To circumvent this, non-gene editing technologies, such as the shRNA-based approach used for Celyad Oncology's CYAD-211, have entered clinical development[35]. While this strategy may improve accessibility, the efficiency and durability of gene knockdown could pose challenges to achieving long-term efficacy.

Fratricide

UCAR-T cells are particularly crucial for treating T cell malignancies, as manufacturing autologous products is often unfeasible due to contamination with malignant T cells. A primary challenge in this context is fratricide, which occurs when CAR T cells target CD7, a common antigen expressed on both malignant and engineered T cells. Fortunately, early studies suggest that CD7 is largely dispensable for T cell development and function, providing a strong rationale for its removal[59, 60]. While gene knockout is the most common method for ablating CD7 expression[33, 46, 47, 61–66], innovative approaches have also emerged. For example, one strategy uses a "natural selection" process during manufacturing to generate CD7-negative CAR T cells (NS7CAR) without additional gene editing. In these cells, the CD7-CAR is thought to sequester the CD7 protein intracellularly, preventing fratricide without requiring a separate blocking molecule. This approach has demonstrated promising safety and efficacy in patients with relapsed/refractory T cell acute lymphoblastic leukemia (T-ALL) or T cell lymphoblastic lymphoma (T-LBL)[61, 67].

Improvements of gene-editing technologies

Both ZFNs and Transcription Activator-Like Effector Nucleases (TALENs) are engineered fusion proteins composed of a customizable DNA-binding domain and a nuclease catalytic domain. While both platforms require re-engineering to target new genomic sites, the modular nature of the TALEN DNA-binding domain simplifies the

design process, making it more flexible and cost-effective. In contrast, ZFNs require more complex engineering and screening and carry a higher risk of off-target effects. Furthermore, an earlier study indicated that the knockout efficiency at the TRAC or TRBC loci using ZFN mRNA electroporation is approximately 20–40%[16]. Consequently, TALENs have been widely adopted for generating UCAR-T cells, with several products such as UCART19, UCART123, and ALLO-501 currently under clinical evaluation[18–21, 57, 58, 68].

The CRISPR/Cas9 system offers significant advantages in simplicity and efficiency, as it uses a guide RNA (gRNA) for target recognition, eliminating the need for complex protein engineering. Its high efficiency for multiplex editing has made it the predominant method for UCAR-T cell manufacturing, with electroporation of Cas9 ribonucleoprotein (RNP) complexes being the primary delivery approach[25, 69]. Additionally, from an industrial perspective, CRISPR has more diverse technical patents, and RNPs are also more amenable to large-scale production than mRNA. However, a key challenge is the potential for off-target effects, which occur when the gRNA binds to partially mismatched genomic sequences[48]. This risk is influenced by target selection and gRNA design, and can be evaluated using computational prediction tools[43]. Furthermore, novel high-fidelity nucleases, such as Cas-CLOVER, have been developed to minimize off-target activity through more stringent target recognition[42].

Conventional gene editing tools like ZFNs, TALENs, and CRISPR function by inducing DNA double-strand breaks (DSBs). A key drawback is that creating multiple DSBs during multiplex editing increases the risk of chromosomal loss or rearrangement[18, 20, 70]. Base editing technology offers a promising alternative that avoids DSBs by fusing a DNA deaminase with a nuclease-impaired Cas protein to facilitate targeted nucleotide conversion[44, 45]. This approach produces precise, predictable edits, in contrast to the stochastic and heterogeneous outcomes of DSB repair pathways[44].

While base editing avoids DSBs, a significant concern is the induction of sgRNA-independent, off-target mutations across the genome. These mutations are driven by the inherent DNA-binding affinity of the deaminase component and have been linked to chromosomal abnormalities[71–73]. Consequently, a key goal has been to engineer base editor variants with minimal off-target activity, and such efforts have proven successful[46, 74, 75]. This improved safety profile offers a distinct advantage over conventional Cas9. For example, base editing achieves high multiplex editing efficiencies (72.0%–96.2%) without compromising T cell yield, whereas spCas9 can reduce yields by up to 41.4% and

induce chromosomal abnormalities in 22% of cells[46]. By preserving genomic stability and avoiding the upregulation of apoptotic pathways, base-edited cells may achieve greater *in vivo* persistence. Consequently, several UCAR-T cell products created with base editing, including BEAM-201, are now entering clinical trials[47]. Furthermore, strategies combining different nuclease platforms are being explored to mitigate chromosomal translocations in UCAR-T cells[45, 70].

CRISPR/Cas9 still plays a dominant role in the current manufacturing of UCAR-T cells (Table 1). The risks of gene editing, such as chromosomal translocations, unintended point mutations, and insertions/deletions, are receiving increasing scrutiny from researchers and regulatory authorities. While base editing has the potential to surpass CRISPR/Cas9 in terms of efficiency, safety, and cell viability, expanding its capabilities to include gene insertion and deletion will be critical for broader application. Ultimately, minimizing off-target activity remains a fundamental challenge common to all gene editing platforms.

Multiple cell types for UCAR-T generation

Several cell populations, including $\gamma\delta$ T cells, invariant natural killer T cells (iNKTs), double-negative T cells (DNTs), and virus-specific T cells (VSTs), are ideal sources for UCAR-T manufacturing due to their inherently low risk of inducing GvHD. Additionally, alternative sources such as induced pluripotent stem cells (iPSCs) and placental circulating T (P-T) cells can be engineered to possess advantageous phenotypes (Fig. 1; Table 2). The unique antigen recognition mechanisms of some of these cell types may also enhance their efficacy against solid tumors.

$\gamma\delta$ T cells, which comprise approximately 5% of peripheral blood CD3+ cells, are predominantly of the V γ 9V δ 2 subtype[98]. Their ability to recognize antigens in an HLA-independent manner minimizes the risk of GvHD. Furthermore, their intrinsic anti-tumor activity enables $\gamma\delta$ CAR T cells to target tumor cells even after loss of the CAR antigen, thereby addressing challenges of low or heterogeneous antigen expression[76]. A leading clinical example is ADI-001, a CD20-targeted $\gamma\delta$ UCAR-T product developed by Adicet Bio[77]. In a Phase I trial in patients with B-cell malignancies, ADI-001 demonstrated a 67% objective response rate (ORR) and complete remission (CR) rate at day 28[99]. Cellular kinetic studies revealed robust, dose-dependent expansion and persistence, irrespective of HLA mismatch[100]. Adicet Bio is now expanding its $\gamma\delta$ CAR T platform to target solid tumors and autoimmune diseases[101].

Given the heterogeneity of solid tumors, allogeneic $\gamma\delta$ T cells, both unmodified and CAR-engineered, are being actively explored as therapeutic options. The adoptive

transfer of V γ 9V δ 2 T cells has already demonstrated a survival benefit in patients with advanced liver and lung cancer, with minimal side effects[98]. The therapeutic advantage of $\gamma\delta$ T cells stems from their dual antigen-recognition mechanisms, which include their native TCR and innate activating receptors like NKG2D. For example, treatment of glioblastoma with temozolomide upregulates the stress ligand NKG2DL, thereby sensitizing tumor cells to $\gamma\delta$ T cell-mediated cytotoxicity[102]. Similarly, a CD123-targeted $\gamma\delta$ UCAR-T product controlled tumor growth for over 70 days in an AML mouse model, an effect associated with enhanced expression of activating NK cell receptors[78]. The V δ 1 T cell subset, which preferentially resides in epithelial tissues, is particularly promising for solid tumors[103]. Adicet Bio is evaluating ADI-270, a CD70-targeted CAR-V δ 1 T cell therapy, in a Phase 1 trial for renal cell carcinoma. This product is armored with a dominant-negative TGF β receptor II (dnTGF β RII) to resist the immunosuppressive TME[104]. Another product, ADI-002, which targets glypican-3 (GPC-3) on hepatocellular carcinoma, has also demonstrated promising preclinical efficacy and safety[80].

Developing robust methods to expand $\gamma\delta$ T cells is crucial for their clinical use. While initial approaches using zoledronic acid often yield products of insufficient purity, newer protocols using novel nitrogen-containing bisphosphonates can achieve superior expansion and purity of V δ 2 T cells[79]. More recently, robust expansion of the V δ 1 subset has been demonstrated by stimulating $\alpha\beta$ TCR- and CD56-depleted PBMCs with an anti-CD3 monoclonal antibody and IL-15[81]. Furthermore, $\gamma\delta$ CAR T cells have been successfully generated from iPSCs, showing potent efficacy when cultured with IL-15[95].

iNKTs are a rare, evolutionarily conserved T cell subset that recognizes glycolipid antigens presented by the monomorphic CD1d molecule. This HLA-independent recognition mechanism makes them an attractive platform for allogeneic therapy, as it confers a low risk of GvHD[105]. iNKT cells can efficiently home to tumors in response to chemokines like CCL2 and CCL20, making them particularly promising for solid tumors[106]. Several preclinical studies have explored CAR-iNKTs for both hematologic malignancies and solid tumors[82, 107]. Currently, the most clinically advanced allogeneic CAR-iNKT cells co-expressing a CD19-specific CAR, IL-15, and shRNA targeting B2M and CD74 to downregulate HLA-I and II molecules, has demonstrated promising efficacy and safety in patients with relapsed or refractory non-Hodgkin's lymphoma (NHL) and acute lymphoblastic leukemia (ALL)[87]. The first-in-human Phase I trial of autologous GD2-CAR iNKT cells co-expressing IL-15 for neuroblastoma has shown early

Table 2 UCAR-T cells generated from alternative cell types

Refs	Transduction	Target	Co-stimulatory domain	Cell type	Cell source	Key cell sorting and stimulation method
[76]	Retrovirus	CD19	CD28	$\gamma\delta$ T	healthy donor PBMC	Zol culture
[77]	Retrovirus	CD20	4-1BB	$\gamma\delta$ T (V δ 1)	healthy donor PBMC	agonistic anti-V δ 1 antibody; FBS + IL-2; MACS
[78]	Retrovirus	CD123	4-1BB	$\gamma\delta$ T (V δ 1)	healthy donor PBMC	MACS
[79]	Retrovirus	CEA	CD28	$\gamma\delta$ T (V γ 9V δ 2)	healthy donor PBMC	PTA + IL-7 + IL-15
[80]	Retrovirus	GPC-3	4-1BB	$\gamma\delta$ T (V δ 1)	healthy donor PBMC	agonistic anti-V δ 1 antibody; FBS + IL-2; MACS
[81]	Retrovirus	B7H3	CD28	$\gamma\delta$ T (V δ 1)	healthy donor PBMC	α BTCT/CD56-depletion by MACS; OKT-3/IL-15 stimulation
[82]	Retrovirus	CD38/BCMA	CD28 or 4-1BB	NKT	healthy donor PBMC	MACS; α GalCer stimulation
[83]	NA	CD19	CD28	NKT	mouse spleen cell	MACS; α GalCer stimulation
[84]	Retrovirus	CD19/B7H3	CD28 or 4-1BB	NKT	mouse spleen cell	MACS; CD3/CD28 mAb stimulation; IL-2/IL-7
[85]	Lentivirus	CD19/BCMA/GD2 et al	4-1BB	NKT	cord-blood CD34 + HSPCs	five-stage Ex Vivo HSPC-Derived NKT Cell Culture
[86]	Lentivirus	BCMA	4-1BB	NKT	cord-blood CD34 + HSPCs	feeder-free HSC differentiation culture
[87]	Retrovirus	CD19	NA	NKT	healthy donor PBMC	NA
[88]	Retrovirus	CD19	CD28 and 4-1BB	NKT	healthy donor PBMC	anti-iNKT MicroBeads; α -GalCer stimulation
[89]	Retrovirus	CD19	CD28	DNT	healthy donor PBMC	CD4 + and CD8 + cell depletion kit; OKT-3 stimulation
[90]	Retrovirus	CD4	CD28 and 4-1BB	DNT	healthy donor PBMC	CD4 + and CD8 + cell depletion kit; OKT-3 stimulation
[91]	Lentivirus	CD19	4-1BB	DNT	healthy donor PBMC	CD4 + and CD8 + cell depletion kit
[39]	Retrovirus	CD30	CD28	VST (EBV)	healthy donor PBMC	CD45RA positive cells depletion by MACS; EBV pepmixes stimulation
[92]	Retrovirus	CD30	CD28	VST (EBV)	healthy donor PBMC	CD45RA positive cells depletion by MACS; EBV pepmixes stimulation
[93]	Lentivirus	CD19	CD28	VST (CMV)	healthy donor PBMC	CMVpp65 stimulation; anti-IFN- γ MACS
[94]	Lentivirus	CD19	CD28	NA	iPSC	Tn/mem reprogramming; PSC-ATO system
[95]	dsDNA HDRT	CD19	NA	$\gamma\delta$ T	iPSC	two-stage culture; IL-2/IL-5 stimulation; Zol + IL-2
[96]	Lentivirus	CD19	NA	NA	iPSC	stroma-free system; G9a/GLP chemical inhibitor
[97]	Retrovirus	CD19	CD28	NA	P-T cells	MACS; serum + IL-2

PBMC peripheral blood mononuclear cell, Zol zoledronate, FBS fetal bovine serum, MACS magnetic-activated cell sorting, CEA the carcinoembryonic antigen, PTA a novel nitrogen-containing bisphosphonate prodrug, GPC-3 glypican-3, TCR T cell receptor, OKT-3 an anti-CD3 monoclonal antibody, NKT natural killer T cell, α GalCer α -galactosylceramide, BCMA B cell maturation antigen, mAb monoclonal antibody, HSPC hematopoietic stem and progenitor cell, NA not available, DNT double-negative T cell, VST virus-specific T cell, EBV Epstein-Barr virus, CMV cytomegalovirus, iPSC induced pluripotent stem cell, PSC-ATO pluripotent stem cells based artificial thymic organoid, HDRT homology-directed repair template, G9a/GLP an H3K9-directed histone methyltransferase, P-T placental circulating T

signs of safety and efficacy, with one complete remission reported among 11 patients[108].

The therapeutic effect of CAR-iNKTs is amplified through complex interactions within the host immune system and TME. In addition to direct CAR- and TCR-mediated cytotoxicity, CAR-iNKTs can induce a broader anti-tumor response by cross-priming host CD8 + T

cells[83]. They can also favorably remodel the TME by depleting immunosuppressive, CD1d-expressing populations such as tumor-associated macrophages (TAMs) and myeloid-derived suppressor cells (MDSCs)[86, 109]. Furthermore, this multifaceted activity can lead to dendritic cell activation, elimination of M2-like macrophages, and

the promotion of epitope spreading to engage an endogenous T cell response[84].

The low frequency of iNKT cells in peripheral blood (<1%) necessitates scalable manufacturing strategies that do not rely on PBMC isolation. Promising approaches include the differentiation of CAR-iNKTs from renewable sources like cord blood-derived hematopoietic stem and progenitor cells (HSPCs) or hematopoietic stem cells (HSCs)[85, 86]. While these platforms require further preclinical validation, a clinically advanced allogeneic CAR-iNKT product is already showing promise..

DNTs, a CD3+subset lacking both CD4 and CD8, are an attractive platform for allogeneic therapy due to their low propensity to cause GvHD. Pioneering work has demonstrated that unmodified allogeneic DNTs can be expanded to clinical scale and safely administered to patients with relapsed AML[110–112]. Their anti-tumor activity, mediated by innate receptors like NKG2D and DNAM-1, has been shown in models of both AML and non-small cell lung cancer[113, 114]. Building on this foundation, CD19-targeted CAR-DNTs have shown efficacy comparable to conventional CAR T cells in pre-clinical models of B-cell leukemia and lung cancer, but without inducing GvHD[89]. The platform's versatility was further shown with CAR4-DNTs, which were effective against T cell malignancies; their persistence was enhanced by the PI3K δ inhibitor idelalisib[90]. These results highlight the potential of CAR-DNTs for treating a variety of hematologic and solid tumors.

A first-in-human Phase I study of an allogeneic CAR-DNT product (RJMty19) in patients with B-cell lymphoma reported a favorable safety profile, with no instances of grade ≥ 3 cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS), GvHD, or dose-limiting toxicities (DLTs). Notably, all three patients in the highest-dose group achieved objective responses[91]. These encouraging results support further investigation, including dose escalation and multi-infusion regimens.

VSTs exhibit a reduced risk of GvHD, a safety feature attributed to their restricted TCR repertoire[115, 116]. Although their generation requires specialized manufacturing, VSTs are an established therapy for viral infections, particularly following hematopoietic stem cell transplantation[117]. However, their direct application as an anti-cancer treatment is currently limited[92, 93].

iPSCs represent a promising renewable source for CAR T cell manufacturing due to their high proliferative capacity and differentiation potential. One strategy involves generating CAR-iPSCs from T cells and subsequently differentiating them into functional CAR T cells using a 3D organoid system. The resulting product

exhibits cytotoxicity comparable to conventional CAR T cells but possesses a more uniform, clonal TCR repertoire and lower MHC expression, potentially reducing the risks of GvHD and graft rejection[94]. another study identified the H3K9-directed histone methyltransferase G9a/GLP as a repressor of T cell lineage commitment. Chemical inhibition of G9a/GLP facilitated the generation of mature iPSC-T cells with transcriptional profiles that closely resemble peripheral blood $\alpha\beta$ T cells. When engineered to express a CAR, these iPSC-derived T cells demonstrated enhanced effector function[96].

The composition of the starting T cell product is a critical determinant of CAR T cell efficacy. A high frequency of T memory stem cells (T_{SCM}) and distinct gene expression profiles within the apheresis cells are strongly associated with durable clinical responses[13]. UCAR-T cells using P-T cells from umbilical cord blood, characterized by their retention of stem cell properties and enhanced uniformity, has shown more favorable cytokine profile, and durable efficacy[97].

Strategies to further improve the efficacy

UCAR-T cells currently face the challenge of poor durability and depth of remission, and researchers hope to improve these cells through different ways to achieve better efficacy (Fig. 2; Table 3). A key approach involves engineering the cells to improve their intrinsic function and persistence. CAR T and UCAR-T face similar challenges in treating solid tumors, including tumor heterogeneity and the TME, which can lead to inadequate efficacy or disease recurrence. Many innovative strategies, such as advanced genetic modifications, are being pioneered to overcome these barriers[118]. These advancements have yet to be widely applied to UCAR-T. Nevertheless, this progress in the autologous field provides a promising roadmap for engineering effective UCAR-T cell therapies for solid tumors.

Intrinsic enhancement of UCAR-T cells

The limited durability and remission depth of UCAR-T cells may stem not only from host immune rejection but also from the genetic modifications required for their production. For instance, while disrupting the endogenous TCR is necessary to prevent GvHD, this can impair T cell functionality. The TCR complex is integral to T cell activation and survival pathways, and its deletion has been shown to decrease CAR T cell persistence in vivo[122, 126], potentially by downregulating genes involved in cell survival and differentiation[25]. Conversely, other genetic edits can have complex effects; knocking out HLA class II molecules, for example, was found to upregulate proliferative signals like CD70 and POLR2L, potentially enhancing cell survival[25].

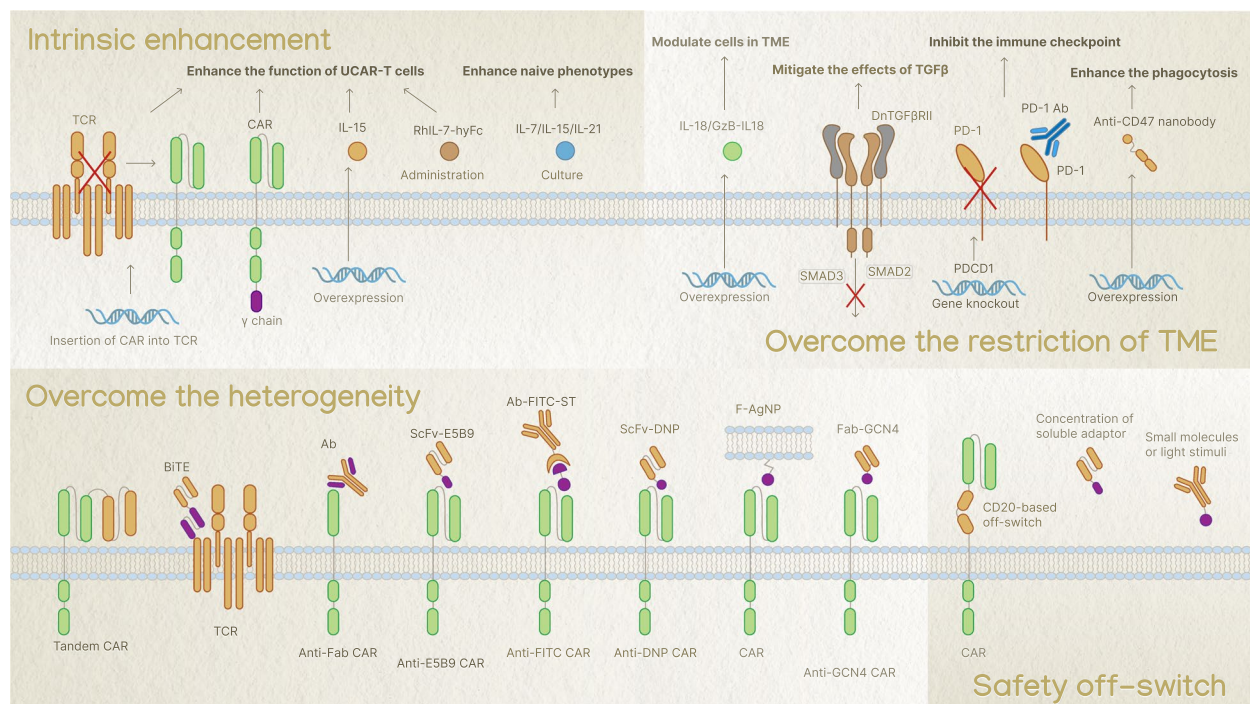


Fig. 2 Possible strategies to improve UCAR-T cells efficacy and safety. Strategies to optimize the inherent function and phenotype of UCAR-T cells include enhancing cell function through the insertion of the CAR construct into the endogenous TCR locus, overexpression of IL-15, or administration of RhIL-7-hyFc. To promote a more persistent "naive" phenotype, cells can be cultured with specific cytokine cocktails (e.g., IL-7/IL-15/IL-21). Approaches to mitigate the immunosuppressive effects of the TME include modulating TME cells by overexpressing cytokines like IL-18/granzyme B-activated human IL-18 (GzB-IL18), mitigating the effects of TGF- β by expressing dominant-negative TGF- β receptor II (dnTGF β RII), inhibiting immune checkpoints through PD-1 gene knockout or PD-1 antibody administration, and enhancing phagocytosis by overexpressing anti-CD47 nanobodies. Strategies to address tumor antigen heterogeneity involve using multi-targeting CARs such as tandem CARs, bispecific T cell Engagers (BiTEs), or "universal adaptor" CAR systems. These adaptor systems include CARs that recognize a common tag (e.g., anti-Fab CAR, anti-E5B9 CAR, anti-FITC CAR, anti-DNP CAR, anti-GCN4 CAR) which can be linked to various tumor-specific antibodies or fusogenic antigen-loaded nanoparticles (F-AgNP), allowing for flexible targeting of diverse antigens. To enhance the safety profile of UCAR-T cells by enabling their controlled elimination or deactivation, strategies include CD20-based off-switches (allowing depletion with rituximab), systems controlled by the concentration of soluble adaptors, or those responsive to external stimuli such as small molecules or light

Therefore, as UCAR-T cells undergo increasingly complex genetic engineering, understanding the functional impact of each modification on fundamental T cell biology is critical.

Most CAR constructs incorporate a co-stimulatory domain derived from either CD28 or 4-1BB, which is critical for T cell survival and function. However, direct comparisons between these domains have been inconclusive, largely due to confounding variables in CAR design across different studies. Preclinical data generally indicate that CD28 signaling yields more robust cytokine production, whereas clinical evidence often suggests that 4-1BB-based CARs exhibit superior persistence. Therefore, to accurately assess the distinct contributions of each domain, future head-to-head comparisons must be conducted using an otherwise identical CAR backbone[127].

Strategies to enhance CAR T cell proliferation and activity, such as providing supplemental cytokine support, are being adapted for UCAR-T platforms[128–131].

This can be achieved by engineering cells to co-express cytokines or through systemic cytokine administration, which can improve in vivo persistence, mitigate exhaustion, and boost anti-tumor effects[78, 80, 85–87, 107, 108]. For instance, while CD2 deletion can impair effector cytokine secretion (e.g., granzyme B, IFN- γ) and reduce cytotoxicity, these deficits can be rescued by administering a long-acting recombinant IL-7 (rhIL-7-hyFc)[119]. Similarly, co-expressing the common cytokine receptor γ -chain (γ c) can augment IL-2 signaling in engineered cells, such as those with CD7 knockout[33]. However, constitutive cytokine activity raises significant safety concerns, including CRS. To circumvent this, novel approaches are being explored, such as a granzyme B-activated IL-18 (GzB-IL18). This molecule is secreted in a latent state and activated only upon T cell engagement, promoting localized anti-tumor activity and myeloid cell reprogramming while minimizing systemic toxicity[121].

Table 3 The strategies of gene knockout, overexpression and co-administration

Refs	Transduction	Target	Co-stimulatory domain	Strategy	Outcomes
[78]	Retrovirus	CD123	4-1BB	IL-15 co-administration	prolong the survival of UCAR-T cells, allowing a single dose to maintain mice free of leukemia
[80]	Retrovirus	GPC-3	4-1BB	IL-15 overexpression	enhance the proliferation ability and therapeutic effect of UCAR-T
[85]	Lentivirus	CD19/BCMA/GD2 et al	CD28 or 4-1BB	IL-15 overexpression	improve the tumor elimination in mice and prolong their survival
[86]	Lentivirus	BCMA	4-1BB	IL-15 overexpression	improve long-term antitumor efficacy
[87]	Retrovirus	CD19	NA	IL-15 overexpression	NA
[119]	Lentivirus	CD2	CD28	rhIL-7-hyFc co-administration	enhance UCAR-Teffector function and prolong the persistence
[120]	Lentivirus	CD19/CD33	4-1BB	rhIL-7-hyFc co-administration	enhance UCAR-T expansion, persistence, and anti-tumor efficacy in vivo
[33]	Retrovirus	CD7	4-1BB	yc overexpression	promote IL-2 production in UCAR-T and enhance in vitro cytotoxicity
[121]	Retrovirus	MUC1	CD28	GzB-IL18 overexpression	promote anti-tumor activity and myeloid cell re-programming
[122]	Lentivirus	MSLN	CD28	PDCD1 knockout by CRISPR/Cas9	enhance cytotoxicity and anti-tumor efficacy in vivo
[123]	AAV	EGFRvIII	4-1BB	PDCD1 knockout by CRISPR/Cas9	prolong survival in mice bearing intracranial tumors
[43]	Lentivirus	CD19	4-1BB	PDCD1 knockout by CRISPR/Cas9	enhance cytotoxicity in vitro
[44]	Lentivirus	CD19	NA	PDCD1 knockout by base editing	demonstrate significant killing of tumor cells overexpressing PD-L1
[46]	Lentivirus	CD7	CD28	PDCD1 knockout by base editing	no significant change in survival
[88]	Retrovirus	CD19	CD28 and 4-1BB	PD-1/PD-L1 inhibitor co-administration	increase cytokine release, cytotoxicity and survival of UCAR-T cells
[124]	Lentivirus	BCMA	4-1BB	CD47-SIRPα blocker overexpression	enhance the phagocytosis of macrophages
[90]	Retrovirus	CD4	CD28 and 4-1BB	PI3Kδ inhibitor	promote memory phenotype and further improve the persistence and function of UCAR-T cells
[125]	mRNA	HLA-G	4-1BB	PD-L1/CD3ε BiTE overexpression	redirect Nb-CAR-γδT and recruit bystander T cells against tumor cells

Abbreviations: GPC-3 glypican-3, BCMA B cell maturation antigen, NA not available, rhIL-7-hyFc a long-acting recombinant human IL-7, yc common cytokine receptor γ chain, MUC1 the transmembrane glycoprotein mucin 1, GzB-IL18 granzyme B-activated human IL-18, MSLN mesothelin, PDCD1 programmed cell death protein 1, CRISPR clustered regularly interspaced short palindromic repeats, Cas9 CRISPR-associated protein 9, AAV adeno-associated virus, EGFRvIII the epidermal growth factor receptor variant III, SIRPα signal regulatory protein alpha, PI3Kδ phosphoinositide 3-kinase δ, HLA human leukocyte antigen, BiTE bispecific T-cell engager

While IL-2 has traditionally been used for in vitro CAR T cell expansion, it primarily drives an effector phenotype[129]. In contrast, culturing cells with cytokines such as IL-7 and IL-15 promotes the generation of less differentiated populations, including T_{SCM} cells, which are associated with superior in vivo expansion and persistence[129, 132, 133]. Beyond cytokine manipulation, advanced manufacturing strategies can preserve this desirable naive state. For example, high-fidelity gene editing systems like Cas-CLOVER enable direct modification of resting T cells, yielding products highly enriched in T_{SCM}[42]. Similarly, targeting CAR integration to the TCR locus can mitigate tonic signaling and prevent cellular exhaustion[23, 24, 134]. Using a drug-regulatable

system to downregulate CAR structure or treating with the multi-kinase inhibitor dasatinib can induce CAR T cells to rest, resulting in the acquisition of a memory-like phenotype and reversal of the exhausted state[135].

Restriction and reshaping of TME

In treating solid tumors, both CAR T and UCAR-T therapies are challenged by the immunosuppressive TME. The TME hinders T cell function through multiple mechanisms. Physical barriers, such as dense stroma and aberrant vasculature, limit the penetration and infiltration of CAR T cells[136]. To address this, CAR T cells have been engineered with synthetic Notch (synNotch) receptors that induce localized secretion of matrix-degrading

enzymes, enhancing tumor access and clearance[137]. Furthermore, poor tumor homing often results from a mismatch between tumor-secreted chemokines and the receptors on T cells. Expressing appropriate chemokine receptors, like CXCR2, on CAR T cells has been shown to significantly improve their trafficking and tumor-targeted infiltration[138, 139].

Second, the TME is populated by immunosuppressive cells that inhibit T cell function and promote tumor progression. These include TAMs, MDSCs, and regulatory T cells (Tregs), which collectively suppress anti-tumor immunity by secreting factors like IL-10 and TGF- β , expressing inhibitory ligands, and directly impairing effector T cell activity[140, 141]. Therefore, strategies targeting these cells or their suppressive mechanisms are critical for enhancing CAR T efficacy[142]. For example, a BCMA-UCAR-T designed to secrete a CD47-SIRP α blocker not only prevents tumor cells from evading macrophage phagocytosis but also shifts the TME towards a more pro-inflammatory state by reducing MDSCs and increasing M1-type macrophages[124]. This may be achieved by reducing myeloid-derived stem cells and increasing CD11c⁺ dendritic cells and M1-type macrophages in tumor tissue[143]. Similarly, CAR-iNKT cells can deplete TAMs and MDSCs via their CD1d-restricted activity[86]. An alternative strategy is to make CAR T cells intrinsically resistant to immunosuppressive signals. Armoring UCAR-T cells, such as V δ 1 T cells, with a dominant-negative TGF- β receptor renders them functional even in a TGF- β -rich environment [104].

The TME also presents a significant metabolic challenge. Highly glycolytic tumor cells create a hypoxic, acidic, and nutrient-poor environment where T cells struggle to compete for energy. This metabolic stress, combined with chronic antigen stimulation and immunosuppressive signals, drives T cells toward exhaustion. This state is also induced by antigen-independent tonic signaling from the CAR construct itself. Exhaustion is characterized by global transcriptional and epigenetic changes, including upregulation of inhibitory receptors (e.g., PD-1, TIM-3), reduced cytokine secretion, and impaired mitochondrial oxidative phosphorylation[144–146]. As previously noted, incorporating a 4-1BB co-stimulatory domain or using specific cytokine combinations can promote more durable T cell phenotypes. Additionally, modulating metabolic pathways like PI3K-AKT-mTOR or directly enhancing mitochondrial function has shown promise in reversing CAR T cell exhaustion[146].

While the aforementioned strategies modify the intrinsic properties of CAR T cells, directly targeting the TME is also crucial. A primary approach involves disrupting the PD-1/PD-L1 axis, a key driver of T cell exhaustion.

This can be achieved with checkpoint inhibitors or, more permanently, through PDCD1 gene knockout, a strategy extensively explored in UCAR-T development[43, 44, 46, 88, 123, 147]., though its clinical application remains limited. Furthermore, immune checkpoints can be repurposed as therapeutic targets. For instance, V δ 2 T cells engineered to express an HLA-G CAR while secreting a PD-L1/CD3 ϵ bispecific T cell engager (BiTE) can effectively eliminate solid tumors expressing either PD-L1 or HLA-G [125].

Finally, engineering CAR T cells to secrete specific cytokines is a powerful strategy for remodeling the TME. For instance, IL-12 can trigger chemokine production (CCL5, CCL2, and CXCL10) to recruit more T cells, while also reducing Tregs and activating myeloid cells[148, 149]. Secreted IL-15 or IL-15R α targeting groups integrated into CAR eliminate MDSCs and reduce the secretion of immunosuppressive molecules[150]. IL-18 or GzB-IL18 can promote localized anti-tumor activity and myeloid cell reprogramming[121]. Conversely, blocking cytokines like IL-4, which epigenetic studies have identified as a driver of exhaustion, can enhance T cell function and persistence[151, 152]. Therefore, combining multiple cytokine modifications holds great promise for improving UCAR-T efficacy, though the risk of off-tumor toxicities from systemic immune activation must be carefully managed.

Tumor heterogeneity

The development of more effective targets, whether tumor-specific or broadly expressed, remains a key objective. Myeloid malignancies are particularly challenging due to a scarcity of specific surface antigens. A recent preclinical study addressed this by using adenine base editing to create UCAR-T cells targeting the pan-hematopoietic marker CD45, demonstrating broad efficacy against various leukemias and lymphomas[153]. Epitope editing on the UCAR-T cells and hematopoietic stem cells preserved CD45 function while preventing fratricide, thereby safeguarding hematopoiesis. This approach suggests a potential strategy for treating a wide range of hematopoietic malignancies by replacing the patient's native hematopoietic system[153]. For solid tumors, antigen heterogeneity and escape are also significant hurdles. Multi-targeting strategies using dual-antigen CARs, BiTEs, or switchable adaptors can improve tumor eradication and mitigate antigen loss. However, enhanced targeting capabilities increase the risk of off-tumor toxicity, demanding more sophisticated safety mechanisms in UCAR-T development.

Tandem CARs, which incorporate two antigen-binding domains into a single construct, can mitigate tumor escape and enhance antitumor activity. This approach has proven effective in glioblastoma models, where CAR T cells co-targeting HER2 and IL13R α 2, or EGFR and IL13R α 2, demonstrated superior efficacy[154, 155]. Similarly, a CD70/B7-H3 tandem CAR improved tumor control in models of lung cancer and melanoma[156]. CAR T cells targeting two antigens (GD2 and B7-H3) simultaneously, combined with trans-CD28 and 4-1BB co-stimulation and sharing the same CD3 ζ chain, effectively prevented tumor escape driven by low antigen density in a neuroblastoma model[157].

BiTEs, which link CD3 on T cells to a tumor-associated antigen (TAA), can be combined with CARs to overcome antigen heterogeneity. When secreted by CAR T cells, BiTEs better direct engineered T cells towards tumor cells, and recruit bystander T cells to attack tumor cells that lack the primary CAR target[158]. This approach can be multifunctional; for example, a BiTE targeting PD-L1 can also provide simultaneous checkpoint blockade[125].

Switchable CAR T platforms allow for retargeting to different antigens without cell remanufacturing, offering significant therapeutic flexibility (Table 4). These systems can overcome pre-existing drug resistance, as demonstrated by adaptor-mediated CAR T cells that successfully targeted cetuximab-resistant tumors[159]. A universal CAR uses a non-targeting peptide that binds to a specific pocket engineered into therapeutic antibodies, allowing specificity to be controlled by the co-administered antibody[160]. The UniCAR platform uses separate target modules (TMs) to bridge the CAR T cell to tumor antigens like CD123 or CD19[161–163]. Conjugation of antibody fusion proteins to FITC-SpyTag (FITC-ST) peptides significantly enhanced the cytotoxicity of anti-FITC CAR T cells against tumor cells[164]. 2,4-dinitrophenol (DNP)-conjugated antibodies have also shown utility

as adaptors for guiding CAR T cells target to different tumor antigens[165]. For all adaptor systems, optimizing the binding affinity between the CAR and the adaptor is critical for balancing antitumor efficacy with toxicity[166, 167]. A different approach artificially decorates tumor cells with new targets. Fusogenic antigen-loaded nanoparticles (F-AgNPs) can fuse with tumor cell membranes to deposit exogenous antigens, creating targets for CAR T cells[168]. Unlike adaptors, this strategy directly addresses low antigen density but faces challenges related to in vivo nanoparticle delivery.

Safety switch in CAR T cell therapy

Incorporating "off-switches" into the CAR design is a critical strategy for managing toxicity and facilitating the timely clearance of CAR T cells. A common approach involves engineering a targetable epitope, such as the CD20 mimotope recognized by rituximab, into the CAR construct. This sensitizes the UCAR-T cells to elimination via complement- or antibody-dependent cytotoxicity, allowing for bone marrow recovery after leukemia clearance without compromising remission, a strategy applied across numerous platforms[19, 21, 22, 44, 57, 81]. Alternatively, activity can be controlled in real-time using adaptor molecules with short half-lives. For instance, UniCAR T cell activity can be rapidly toggled by managing the infusion of a CD123-specific targeting module (TM123)[162, 163]. Similarly, tumor targeting compounds obtained from DNA encoded compound libraries bound to 1, 3-dione hapten, as a small molecule bifunctional linker, can transiently bridge CAR T cells to tumors[169]. Furthermore, adaptors can be engineered to respond to external stimuli like small molecules or light, providing additional layers of conditional control over UCAR-T cell activity[170].

Table 4 Switchable adaptor designs in CAR T cells

Refs	Transduction	Target	Co-stimulatory domain	adaptor
[159]	Retrovirus	CD16 158 V	CD28	cetuximab
[160]	Lentivirus	Fab epitope	CD28 or 4-1BB	modified HER2 IgG and Fab
[161]	Lentivirus	E5B9 epitope	CD28	anti-CD10 target module containing epitope E5B9
[162, 163]	Lentivirus	E5B9 epitope	CD28	anti-CD123 target module containing epitope E5B9
[164]	Lentivirus	FITC	CD28	antibody labelled by FITC
[165]	Lentivirus	DNP	CD28	anti-CD19 scFv with site-specific dinitrophenylation
[166]	Lentivirus	GCN4 epitope	4-1BB	anti-ROR1 Fab fused with GCN4 peptide-tagged Fab
[167]	Lentivirus	CD16a	CD28, 4-1BB and CD247	IgG1
[168]	NA	EGFRvIII	4-1BB	F-AgNPs mediated cell membrane antigen modification and transfer

Abbreviations: HER2 human epidermal growth factor receptor 2, IgG immunoglobulin, Fab antigen-binding fragment, FITC fluorescein isothiocyanate, DNP 2,4-dinitrophenol, scFv single chain fragment variable, GCN4 a eukaryotic transcriptional activator protein, ROR1 receptor tyrosine kinase like orphan receptor 1, NA not available, EGFRvIII the epidermal growth factor receptor variant III, F-AgNP fusogenic antigen loaded nanoparticle

Trends in clinical trials

Using the Trialtrave database, we identified 169 clinical trials with UCAR-T cells as the initial drug as of 31 October 2024. The results are depicted in Fig. 3.

Analysis of 169 UCAR-T clinical trials reveals a significant increase in trial initiations from 2016 to 2023 (Mann–Kendall test, $p=0.042$), with a peak increase in 2019. China leads this landscape with 75 trials (44%), followed by the United States with 54 (32%). The vast majority of these studies are in early stages: 131 (78%) are Phase

I and 24 (14%) are Phase I/II. (two Phase IV studies were excluded as long-term follow-up of existing studies).

CD19 remains the dominant target (Fig. 3D), although other antigens such as CD7, CD22, BCMA, and CD20 are also frequently pursued. Development efforts are also exploring novel targets, including CD276, CD33, and CD70. Consistent with the success of autologous CAR T therapies, the clinical landscape is heavily focused on hematological malignancies, particularly non-Hodgkin's lymphoma and acute lymphocytic leukemia (Fig. 3C). In



Fig. 3 Trends in clinical trials. **A** Number of UCAR-T clinical trials by year (2016–2024). **B** Number of UCAR-T clinical trials in different countries (top 15). **C** Number of UCAR-T clinical trials for different tumor species. **D** Number of UCAR-T clinical trials by different targets

contrast, trials in solid tumors are less common, though they span a range of indications including colorectal, renal, and ovarian cancers.

Regulatory hurdles and health economic challenges

As gene-edited allogeneic cell products, the clinical translation and widespread application of UCAR-T therapies are overseen by global regulatory agencies. In the United States, the FDA's guidance, "Considerations for the Development of Chimeric Antigen Receptor (CAR) T Cell Products," outlines regulatory requirements to mitigate the risks of UCAR-T therapies[171]. This document delineates foundational requirements applicable to all CAR T modalities, including standards for product characterization (CAR construct and vector), Chemistry, Manufacturing, and Controls (CMC), and nonclinical and clinical safety evaluations, including the management of toxicities like CRS and neurotoxicity. Critically for UCAR-T products, the guidance addresses the distinct challenges posed by allogeneic therapies. The guidance mandates that clinical protocols mitigate key additional risks, such as GvHD, through detailed immunological matching strategies and a comprehensive GvHD management plan. This plan must integrate specific monitoring and grading systems and be formally factored into the study's safety endpoints, including the definitions of DLTs and study stopping rules.

The European Medicines Agency (EMA) follows a similar risk-based approach, classifying UCAR-T products as Advanced Therapy Medicinal Products (ATMPs). "Guideline on quality, non-clinical and clinical aspects of medicinal products containing genetically modified cells" provides a comprehensive framework for development[172]. While this guideline covers foundational requirements for quality, safety, and efficacy common to all cell-based gene therapies, it places significant emphasis on the specific technologies underpinning the "universal" nature of UCAR-T cells. In this context, the guideline addresses the use of genome editing techniques employed to disrupt the endogenous TCR and prevent GvHD. Consequently, it mandates a rigorous assessment of potential genomic alterations, requiring developers to characterize the specificity and safety of the editing process itself. Furthermore, the EMA framework distinctly considers the dual challenge of allogeneic immunogenicity: not only the risk of GvHD, but also the potential for HvGR. This technology-centric focus requires a deep, mechanistic understanding of the product to be established throughout the non-clinical and clinical development phases.

In China, the National Medical Products Administration (NMPA) has established its regulatory pathway

through a series of technical guiding principles[173–175]. A distinct feature is the focus on the non-clinical evaluation of the gene-editing technologies central to UCAR-T products, mandating rigorous assessment of on- and off-target effects. Clinically, the NMPA emphasizes the unique cellular kinetics of these "living drugs" and, notably, specifies a long-term follow-up of at least 15 years for products utilizing integrating vectors to monitor for delayed adverse events. These guiding principles underscore a regulatory emphasis on both the mechanistic underpinnings and the long-term safety profile of UCAR-T therapies.

For autologous CAR T cell therapy, high cost is the primary barrier to access [176]. UCAR-T therapies hold the promise of democratizing access by enabling economies of scale. Centralized manufacturing of batches from healthy donors could theoretically lower the per-patient cost compared to autologous products and dramatically shorten the critical vein-to-vein time for patient. However, the upfront investment required for developing proprietary gene-editing technologies, establishing and maintaining GMP-compliant master cell banks, and conducting the safety and comparability studies mandated by regulatory agencies may add additional costs. Therefore, realizing the transformative potential of UCAR-T therapy will require a concerted, multi-stakeholder effort to innovate manufacturing and analytical technologies to reduce costs.

Efficacy of UCAR-T cell therapy in hematologic malignancy and solid tumor

Clinical trials of UCAR-T cell therapy with published or interim results are delineated in Table 5, illustrating both significant advancements and persistent challenges in the field. A key insight is the encouraging efficacy observed in hematological malignancies, where several early-phase studies targeting CD7 and CD19 have reported high initial response rates. In contrast, a significant challenge remains in treating solid tumors, with therapies demonstrating more limited activity. It is also important to note that these promising outcomes are largely derived from early-phase studies, reflecting the evolving nature of the field and the need for further research to optimize and broaden the applicability of these therapies.

These promising results in hematologic malignancies are exemplified by several recent trials, targeting antigens such as CD19, CD22, CD70, BCMA, CD5, and CD7. Recently, Frederick L Locke et al. conducted a phase 1 trial, which evaluated allogeneic CD19 CAR T cells (cema-cel/ALLO-501) in 33 CD19 CAR T-naïve patients with relapsed/refractory large B-cell lymphoma[177]. Following lymphodepletion with fludarabine and cyclophosphamide (FC) and ALLO-647, CAR

Table 5 UCAR-T cells with published clinical experience

Target	Year	Drug	Tumor type	Cell Source	Feature	Dose	Multiple doses	Lymphodepletion	Other treatment	Response	Phase	NCT	Refs
B7-H3	2023	MT-027	rHGG	Healthy donor	CRISPR disruption of TCR and HLA-I, intra-lumbar injection every month	2.5×10^7	Yes	NA	NA	ORR 42.9%	I	ChiCTR2100047968	[182]
BCMA	2023	P-BCMA-ALL01	R/R MM	Healthy donor	T stem cell memory (TSCM)-rich, knockout of TRBC and B2M	2×10^6 /kg	Single or multiple	F 30 mg/m ² + C 300, 500, 1000 mg/m ² × 3	NA	ORR 82%	I	NCT04960579	[183]
BCMA	2023	CYAD-211	MM	Healthy donor	shRNA knockdown of CD3ζ at the mRNA level	30–100 × 10 ⁶	No	F 30 mg/m ² + C 300 mg/m ² × 3	NA	ORR 25%	I	NCT04613557	[184]
BCMA	2023	ALLO-715	R/R MM	Healthy donor	TALEN disruption of TRAC and CD52	40–480 × 10 ⁶	No	F 30 mg/m ² + C 300 mg/m ² × 3 + ALLO-647	NA	ORR 70.8% CR 25%	I	NCT04093596	[21]
CD123	2023	UCART123	R/R AML	Healthy donor	TALEN disruption of TCRαβ and CD52	0.125–3.3 × 10 ⁶ /kg	No	F 30 mg/m ² × 4 + C 750 mg/m ² × 3 ± A 12 mg × 4	NA	FCA arm at DL2: 12.5% CR	I	NCT03190278	[19]
CD19	2025	Cemacabtagene Ansegtedileucel/ALLO-501	R/R LBCL	Healthy donor	TALEN disruption of TRAC and CD52	120×10^6	Single or multiple	F 30 mg/m ² + C 300 or 500 mg/m ² × 3 + ALLO-647	NA	ORR 58%, CR 42%, mDoR 23:1 m in CR	I/II	NCT04416984	[177]
CD19	2024	donor-derived CD19 CAR T	HR-B-ALL	Healthy donors	Allo-HSCT donors, prophylactic use	1.0–4.9 × 10 ⁶ /kg	No	F 25 mg/m ² + C 250 mg/m ² × 3	allo-HSCT	2 yr PFS 84%	II	ChiCTR 2000041025, ChiCTR-ONN-16009862	[185]
CD19	2023	GC007g	R/R B-ALL	allo-HSCT donor	HSCT donor-derived CAR T cells	6–20 × 10 ⁵ /kg	No	F 30 mg/m ² + C 300 mg/m ² × 3	NA	12 m ORR 75%	I	NCT04516551	[186]
CD19	2023	anti-CD19 EBV-CTL	R/R B-cell malignancies	EBV-specific cytotoxic lymphocytes	CD28-containing CAR	1–10 × 10 ⁶ /kg	Yes	NA	C 3000 mg/m ²	24 m OS 74%	I	NCT01430390	[187]
CD19	2023	ThisCART19	R/R B-ALL	Healthy donor	Retention of TCRαβ/CD3 complex	3–5 × 10 ⁶ /kg	No	F 30 mg/m ² + C 300 mg/m ² × 3	NA	CR 100%	I	NCT05350787	[188]
CD19	2023	RJMty19	R/R LBCL	Healthy donor	Double negative T cell	1–20 × 10 ⁶ /kg	No	F 25 mg/m ² + C 500 mg/m ² × 3	NA	ORR 25% CR 8.3%	I	NCT05453669	[189]

Table 5 (continued)

Target	Year	Drug	Tumor type	Cell Source	Feature	Dose	Multiple doses	Lymphodepletion	Other treatment	Response	Phase	NCT	Refs
CD19	2022	CTX110	R/R LBCL	Healthy donor	CRISPR disruption of TRAC and β_2M	30–600 × 10 ⁶	Single or multiple	F 30 mg/m ² + C 500 mg/m ² × 3	NA	≥ 300 × 10 ⁶ ; ORR 67% CR 41%	I	NCT04035434	[190]
CD19	2022	UCART19	R/R B-ALL	Healthy donor	TALEN disruption of TRAC and CD52	30–240 × 10 ⁶	Single or multiple	F 30 mg/m ² + C 500 mg/m ² × 3 + A 1 mg/kg/40 mg/60 mg	NA	ORR 48%	I	NCT02746952	[58]
CD19	2021	KUR-502	R/R B-cell malignancies	Healthy donor	NKT cells	10–100 × 10 ⁶ /m ²	No	F 30 mg/m ² + C 500 mg/m ² × 3	NA	ORR 80%	I	NCT03774654	[87]
CD19	2020	anti-CD19 CAR T cell	R/R B-ALL	allo-HSCT donor	HSCT donor-derived CAR T cells	5–10 × 10 ⁶ /kg	No	F 30 mg/m ² + C 300 mg/m ² × 3	NA	CR 72.7%	I	NCT03275493	[191]
CD19	2016	CAR19 T cell	R/R B-cell malignancies	allo-HSCT donor	HSCT donor-derived CAR T cells	1–10 × 10 ⁶ /kg	No	No	NA	CR 30%	I	NA	[192]
CD19	2013	CD19 CAR-VST	R/R B-cell malignancies	allo-HSCT donor	HSCT donor-derived virus-specific T cells (CMV, EBV, and AdV-specific)	15–120 × 10 ⁶ /m ²	Single or multiple	No	NA	CR 16.7%	I	NCT00840853	[193]
CD19	2013	CD19 CAR T	R/R B-cell malignancies	allo-HSCT donor	HSCT donor-derived CAR T cells	1–100 × 10 ⁶ /kg	No	No	NA	CR 10%	I	NCT01087294	[194]
CD19, CD7	2022	GC502	R/R B-ALL	Healthy donor	CRISPR disruption of TRAC and CD7	10–15 × 10 ⁶ /kg	No	F + C	NA	ORR 100% CR 75%	I	NCT05105867	[195]
CD19, CD22	2021	CTA101	R/R B-ALL	Healthy donor	CRISPR/Cas9-disrupted TRAC and CD52	1–3 × 10	No	F 30 mg/m ² + C 1000 mg/m ² × 3 + A 0.2 mg/kg × 5	NA	CR 83.3%	I	NCT04227015	[196]
CD20, CD22	2023	UCART20 × 22	R/R NHL	Healthy donor	TALEN disruption of TRAC and CD52	50–450 × 10 ⁶	No	F 30 mg/m ² + C 500 mg/m ² × 3 + A 12 mg d1, 24 mg d2, d3	NA	ORR 100% CR 67%	I	NCT05607420	[197]
CD22	2023	UCART22	R/R B-ALL	Healthy donor	TALEN disruption of TRAC and CD52	1–5 × 10 ⁶ /kg	No	F 30 mg/m ² + C 0.5 g/m ² + A 20 mg × 3	NA	ORR 67%	I	NCT04150497	[198]
CD30	2023	CD30 CAR EBVST	HL	Healthy donor	EBV-specific T cell	4–80 × 10 ⁷	No	F + C	NA	ORR 75%, CR 37.5%	I	NCT04288726	[199]

Table 5 (continued)

Target	Year	Drug	Tumor type	Cell Source	Feature	Dose	Multiple doses	Lymphodepletion	Other treatment	Response	Phase	NCT	Refs
CD5	2024	CD5 CART cells	R/R T-ALL	Previous SCT donors or newly matched donors	Disrupted CD5 genes	0.5–2 × 10 ⁶ /kg	No	F 30 mg/m ² + C 30 or 300 mg/m ² × 3	NA	100% CR at day 28	I	NCT05032599	[179]
CD7	2024	CD7 CART cells	CD7 ⁺ malignancies	Haploidentical HSCT donor or healthy donor	CRISPR disruption of CD7, TCR/CD3, and RFX5, introducing Nki and yc receptor	2–5 × 10 ⁶ /kg	No	F 30 mg/m ² + C 300 mg/m ² + E 100 mg × 5	Sequential allo-HSCT	CR 100% CR at 1 yr 60%	I	NCT04599556, NCT04538599	[200]
CD7	2023	WU-CART-007	R/R T-ALL/ LBL	Healthy donor	CRISPR deletion of CD7 and TCR	100–900 × 10 ⁶	No	F 30 mg/m ² + C 500 mg/m ² × 3 OR F 30 mg/m ² × 4 + C 1000 mg/m ² × 3	NA	ORR 78% CR 67%	I/II	NCT04984356	[201]
CD7	2023	GC027	R/R T-ALL/ LBL	Healthy donor	CRISPR disruption of CD7 and TCR	4.3–15 × 10 ⁶ /kg	Single or multiple	F + C ± E ± M ± P	NA	CR 91.7%	I	ChiCTR1900025311	[64, 65]
CD7	2021	CD7 CART cells	R/R T-ALL/ LBL	Healthy donor	HLA-matched or haploidentical donors	0.5–1 × 10 ⁶ /kg	No	F 30 mg/m ² + C 250 mg/m ² × 3 OR F 30 mg/m ² + C 30 mg/kg × 3	NA	CR 90%	I	NCT04823091	[202]
CD7	2022	RD13	CD7 ⁺ malignancies	Healthy donor	CRISPR disruption of CD7, TCR/CD3, and RFX5, introducing Nki and yc receptor	10–30 × 10 ⁶ /kg	No	F 30 mg/m ² + C 300 mg/m ² + E 100 mg × 5	NA	ORR 81.8% CR 63.6%	I	NCT04538599	[33]
CD70	2025	CTX130	R/R T cell Malignancies	Healthy donor	CRISPR disruption of TRAC, β2M and CD70	30–900 × 10 ⁶	Single or multiple	F 30 mg/m ² + C 500 mg/m ² × 3	NA	ORR: 46.2%	I	NCT04502446	[178]
CD70	2024	CTX130	ccRCC	Healthy donor	CRISPR disruption of TRAC, β2M and CD70	30–900 × 10 ⁶	Single or multiple	F 30 mg/m ² + C 500 mg/m ² × 3	NA	ORR 6% CR 6%	I	NCT04438083	[180]
GD2	2025	ALLO_GD2-CART01	R/R HR-NB	HLA-matched or an haploidentical donor	iC9 suicide gene	2.4–4.76 × 10 ⁹	No	F 20–25 mg/m ² + C 500–1000 mg/m ² × 3	HSCT	2/5 CR, 1/5 PR	I	NCT03373097	[181]

Table 5 (continued)

Target	Year	Drug	Tumor type	Cell Source	Feature	Dose	Multiple doses	Lymphodepletion	Other treatment	Response	Phase	NCT	Refs
IL13Ra2	2022	GRm13Z40-2	rGBM	Healthy donor	ZFNs disruption of glucocorticoid receptor (GR) (GRm13Z40-2), infused via the Rickham catheter	10 ⁸	4 cycles	NA	Intracranial administration of rhIL-2, dexamethasone	0	I	NCT01082926	[203]
NKG2D	2021	CYAD-101	mCRC	Healthy donor	Co-expression of TIM peptide	100–1000 × 10 ⁶	Yes	No	FOLFOX	ORR 13%	I	NCT03692429	[204]

Abbreviations: R/R relapsed or refractory, HR-NB high-risk neuroblastoma, HR-B-ALL high-risk B-cell acute lymphoblastic leukemia, T-ALL/LBL T cell acute lymphoblastic leukemia/lymphoblastic lymphoma, MM multiple myeloma, B-NHL B-cell non-Hodgkin's lymphoma, HL Hodgkin lymphoma, rHGGs recurrent high-grade gliomas, rGBM recurrent glioblastoma, cRCC clear cell renal cell carcinoma, NK/NK cell inhibitory receptor, mCRC unresectable metastatic colorectal cancer, PTCL peripheral T cell lymphoma, CTCL cutaneous T cell lymphoma, allo-HSCT allogeneic hematopoietic stem cell transplantation, F fludarabine, C cyclophosphamide, A alemtuzumab, E etoposide, M melphalan, P prednisone, ALLO-647 anti-CD52 monoclonal antibody, rhIL-2 recombinant human IL-2

T expansion and persistence up to 4 months were observed. ORR was 58%, with CR in 42% patients; median CR duration was 23.1 months[177]. Hematologic toxicities were common, but no grade ≥ 3 CRS, ICANS, or GvHD occurred[177]. Similarly, CD70-targeted CTX130 in T cell lymphoma showed 46.2% ORR (19.4% CR), though response durability was limited (median 2.5 months for peripheral T cell lymphoma) [178]. Dose escalation ($\geq 3 \times 10^8$ cells) improved efficacy (ORR 51.6%)[178].

Allogeneic CD5/CD7 CAR T therapies also demonstrate significant promise in T cell malignancies. In a phase I study, allogeneic CD7 CAR T (GC027) induced rapid remission (11/12 CR within 1 month), but long-term efficacy remains unconfirmed due to early CAR T clearance post-infusion[64, 65]. Another phase I trial evaluated allogeneic CD5-specific CAR T therapy in 19 relapsed/refractory T-ALL patients, predominantly CD7 CAR T failures[179]. Infusions (10/16 at 1×10^6 cells/kg) induced CR or CR with incomplete count recovery in all patients by day 30[179]. However, durability remains a hurdle, as only four of these patients proceeded to transplant, and just two of the remaining twelve were still in remission at a 14.3-month follow-up [179].

In contrast to the findings in hematologic cancers, the clinical data from solid tumor trials underscore the challenges mentioned previously. The same CTX130 therapy that was effective in lymphoma showed only a 6% ORR in metastatic clear cell renal cell carcinoma (ccRCC), with a median progression-free survival (PFS) of 2.9 months [180]. Recently, a phase I study evaluated allogeneic GD2-specific CAR T cells (ALLO_GD2-CART01) in five children with relapsed/refractory high-risk neuroblastoma, including four with prior allogeneic hematopoietic stem cell transplantation (allo-HSCT)[181]. Partial and complete responses were observed in four patients, including two sustained CRs[181]. Notably, allogeneic CAR T cells showed enhanced proliferation compared to autologous counterparts[181]. Despite disease progression in three cases, prolonged remission (6+ months) was observed in one patient[181]. In solid tumors, the overall modest response rates and variable persistence highlight the necessity for novel engineering strategies to overcome their immunosuppressive microenvironments.

Clinical strategies to improve efficacy and safety

Clinical studies of UCAR-T cell therapy are not limited to the cell product itself, but also involve evaluating the potential value of multiple application strategies (Fig. 2). Reviewing these clinical trial results may influence future drug approval and clinical decisions by clinicians.

Lymphodepletion strategy

Most UCAR-T studies have utilized a lymphodepletion strategy to clear immunosuppressive cells from the microenvironment, increase cytokine availability, and enhance TME modulation, thus facilitating CAR T cell expansion and persistence. Unlike autologous CAR T therapy, UCAR-T cells generally exhibit reduced persistence due to host rejection; therefore, to enhance the expansion and durability of UCAR-T cells, some clinical trials have employed enhanced lymphodepletion[19, 21, 33, 58, 64, 65, 197, 198, 201, 205]. These enhanced regimens may include higher doses of fludarabine and cyclophosphamide than standard autologous CAR T therapy, with some studies including etoposide as an additional lymphodepletion agent. A comparative analysis of two phase I clinical trials evaluating P-BCMA-ALLO1, targeting BCMA for multiple myeloma, and P-MUC1C-ALLO1, targeting MUC1-C for epithelial-derived solid tumors, investigated the impact of varying cyclophosphamide doses on lymphodepletion characteristics and CAR T cell kinetics[206]. The study revealed that baseline white blood cell (WBC) counts were significantly higher in solid tumor patients compared to multiple myeloma patients[206]. For P-BCMA-ALLO1, increasing the cyclophosphamide dose from 300 mg/m² to 500 mg/m² or 1000 mg/m² enhanced lymphodepletion depth and improved CAR T cell expansion[206]. In contrast, P-MUC1C-ALLO1 patients did not exhibit significant improvements in lymphodepletion depth or CAR T cell expansion with a cyclophosphamide dose of 500 mg/m². However, a dose of 1000 mg/m² demonstrated trends toward improved outcomes in both parameters[206]. Notably, the distinct lymphodepletion requirements between these patient populations are unlikely attributable to differences in CAR T cell product attributes, as both therapies are manufactured using identical transposon, gene editing, and donor sourcing techniques[206]. These preliminary results suggest that higher doses of cyclophosphamide may be necessary to achieve effective lymphodepletion and promote CAR T cell expansion in solid tumor patients.

Several studies have incorporated the CD52 monoclonal antibody, combined with CD52^{-/-} UCAR-T cells, an approved treatment for chronic lymphocytic leukemia (CLL), into the lymphodepletion regimen[207]. CD52 is a glycoprotein on various host immune cells, including lymphocytes, monocytes, macrophages, eosinophils, and dendritic cells [207]. CD52⁺ cells can be effectively depleted by anti-CD52 antibodies, extending the lymphodepletion period, and thereby promoting CAR T cell expansion and sustained persistence[21].

The adverse events of intensified lymphodepletion, including stronger bone marrow suppression and

infection risks associated with CD52 monoclonal antibodies, necessitate careful consideration. In one clinical trial using CD52 monoclonal antibody for lymphodepletion, 33% of patients reported activation of cytomegalovirus (CMV), although there were no cases of grade 4 or grade 5 CMV were reported[21]. Consequently, some researchers recommend the use of prophylactic antiviral medications and antibiotics for enhanced care[65]. The decision to implement enhanced lymphodepletion mandates a thorough benefit-risk evaluation based on the individual's baseline health status and the specific cell therapy being utilized.

Broadening the administration routes: intrathecal and intracavitary drug delivery

For intracranial tumors, the presence of the blood–brain barrier poses a challenge for the effective delivery of CAR T cells into the brain via peripheral blood administration. The intrathecal injection may be a viable local treatment method. Several autologous and allogeneic studies have explored intrathecal injection of CAR T cells for treating recurrent GBM (rGBM) in clinical trials [182, 203, 208]. In a clinical trial of autologous CAR T cells for rGBM, six patients received CART-EGFR-IL13R α 2 cells via Ommaya reservoir intrathecal administration. While all six participants demonstrated a reduction in tumor enhancement, they did not meet the ORR criteria[155]. This provides proof of concept for intrathecal injection treating central nervous system tumors.

Recent clinical investigations have delved into the use of UCAR-T cell therapy via intrathecal administration. GRm13Z40-2, a steroid-resistant, IL13R α 2-specific CAR T cell derived from healthy donors, was administered intrathecally via a Rickham catheter, combined with intracranial injection of recombinant human IL-2 and systemic application of dexamethasone [203]. Although no patients showed remission, two individuals exhibited indications of tumor necrosis. Another UCAR-T cell therapy, MT-027, derived from a healthy donor, was administered intrathecally on a monthly basis for recurrent high-grade gliomas (rHGGs). Among the seven patients treated, three showed a positive response to the therapy, with an OS rate of 85.7% at 12 months [182].

The persistence of UCAR-T cells after intrathecal injection remains to be observed. In the clinical trial with GRm13Z40-2 cells, only a few patients had CAR T cells persist beyond 10 weeks post-last infusion, while MT-027 CAR DNA copy numbers persisted for at least 1 month in the cerebrospinal fluid[182]. The survival characteristics of UCAR-T cells in cerebrospinal fluid may relate to the cell design and the immunological microenvironment, requiring further exploration to elucidate these mechanisms.

Compared to systemic administration, intrathecal injection of CAR T cells presents unique safety signals, with notable cell therapy-related adverse effects within the central nervous system. In the clinical trial with autologous CART-EGFR-IL13R α 2 cells, all six patients experienced early and moderate-severe neurotoxicity featuring elements of both ICANS and tumor inflammation-associated neurotoxicity (TIAN)[155]. Additionally, 2 out of 6 patients in this therapy experienced pseudoprogression[155]. After the MT-027 injection, the most common adverse reactions were headache (7/7, 100%) and fever (6/7, 86%)[182].

Furthermore, intracavitary administration, including into the abdominal and thoracic cavities, is a potentially viable mode of administration. Studies exploring intrathoracic injection of UCAR-T cells have been conducted, although their specific efficacy still requires further observation.

Redosing

Due to the large production capacity of UCAR-T cells, it is feasible to employ designs that incorporate multiple administrations. Additionally, the short half-life of UCAR-T cells suggests that repeated administrations could potentially extend their persistence in the body based on pharmacokinetic (PK) characteristics. Several early-phase clinical trials have attempted multiple administrations of UCAR-T cells[58, 64, 65, 180, 182, 183, 190, 193, 203–205, 209]. However, there remains insufficient data to definitively determine whether multiple administrations can improve response rate or duration of response. In a phase I clinical trial of CD70-targeted UCAR-T cell therapy, six patients received one or more repeated infusions. Of these, four patients assessed as having stable disease (SD) prior to the second infusion remained classified as SD after the subsequent infusion, while two patients evaluated as having progressive disease (PD) before their second infusion continued to be assessed as PD afterward[180].

HLA matching

HLA matching may be an effective way to overcome HvGR. CB-010 is a CD19-targeted UCAR-T cell therapy that utilizes PD-1 gene knockout technology and incorporates a 4-1BB costimulatory domain[210]. This therapy is produced utilizing Cas9 CRISPR hybrid RNA-DNA (chrDNA) technology, achieving precise triple genome editing[210]. Clinical studies have shown that patients receiving CB-010 treatment from donors with a minimum of four matched HLA alleles exhibit extended PFS[211]. Additionally, pharmacokinetic data indicate that the expansion and persistence of CB-010 cells are influenced by the degree of HLA matching[211].

However, adopting HLA matching inevitably narrows the eligible patient population for UCAR-T cells. To maintain the convenience of off-the-shelf cell therapy, establishing a bank of UCAR-T cells with different HLA types, rather than using CAR T cells of a single HLA type, could offer a potential solution. However, this approach necessitates the upfront generation of a substantial quantity of CAR T cells from multiple donors. In essence, while HLA matching may lead to increased preparation costs, it is associated with enhanced treatment responses.

Combination therapy

As an off-the-shelf product, UCAR-T cells hold promise for synergistic therapeutic outcomes when combined with other modalities. As previously mentioned, UCAR-T cell therapy has demonstrated promising remission rates in hematologic malignancies; however, it is associated with a relatively high relapse rate. The concurrent use of allo-HSCT offers a novel strategy to address this issue. This combined approach may not only reduce the risk of relapse following allo-HSCT, but also help resolve the challenge of CAR T cell sourcing, while potentially mitigating the risks of GvHD or HvGR. For patients undergoing allo-HSCT, HSCT donors represent potential sources of CAR T cells. There are three potential time points for CAR T cell administration: pre-allo-HSCT, post-allo-HSCT as consolidation therapy, and post-relapse treatment. Recently, a study by Hu et al. evaluated sequential donor-derived or universal CD7 CAR T cell therapy and haploidentical HSCT for treating relapsed or refractory CD7-positive leukemia or lymphoma[200]. All 10 patients achieved a CR, with 6 maintaining minimal residual disease (MRD)-negative CR for over one year[200]. However, the regimen was associated with significant toxicity, including grade 4 pancytopenia in all patients and two deaths from septic shock[200]. This strategy offers a novel alternative for patients with CD7-positive tumors who are ineligible for conventional allogeneic HSCT.

In a different approach, a phase II trial evaluated prophylactic donor-derived CD19 CAR T cell infusion in 23 high-risk B-cell acute lymphoblastic leukemia (B-ALL) patients post-allo-HSCT, showing a significantly reduced 2-year cumulative relapse rate (5.6% vs. 28.8% in controls, $P=0.026$) and favorable PFS (84.0% vs. 57.3%, $P=0.042$)[185]. Treatment-related adverse events included grade 1–2 CRS (47.8%) and manageable hematologic toxicity, with comparable 2-year non-relapse mortality (10.3% vs. 14.4%). Notably, seven Ph⁺ ALL patients maintained durable remission without tyrosine kinase inhibitors[185]. Despite limitations of small sample size and retrospective design, this strategy demonstrates efficacy in reducing relapse risk, supporting

further prospective investigation as a prophylactic therapy for high-risk B-ALL post-allo-HSCT.

In the realm of solid tumors, investigational research is underway to evaluate the potential synergistic effects of UCAR-T cell therapy in conjunction with other treatment modalities. In a recent study, patients with relapsed/refractory metastatic colorectal cancer (mCRC) received FOLFOX standard cycles as preconditioning chemotherapy, followed by CYAD-101 infusions, with 2 out of 15 patients achieving a PR and no DLT or GvHD found[204].

While clinical data on combination strategies for UCAR-T cells are still emerging, the extensive experience with autologous CAR T therapy offers a valuable blueprint for future directions[212–214]. These strategies aim to overcome resistance mechanisms such as T cell exhaustion and antigen escape, which are challenges for both autologous and allogeneic platforms. One of the promising approaches involves creating dual-targeting pressure to prevent antigen-loss relapse. For hematologic malignancies, combining CD19-targeted CAR T cells with CD20-targeting agents like rituximab or bispecific T cell engagers has shown benefits in autologous settings, leading to higher rates of complete remission and improved overall survival[215]. Looking forward, more innovative partnerships are being explored. For instance, the use of cancer vaccines designed to amplify the CAR T cell population in vivo after infusion has shown encouraging preliminary efficacy in solid tumors[216]. For UCAR-T cells specifically, these combinations may not only bolster anti-tumor activity but could also help modulate the host immune system to support the persistence of the allogeneic product.

However, the integration of UCAR-T cells with other treatments requires a careful balancing of benefits and risks. On one hand, UCAR-T cells may synergistically enhance therapeutic effects in combination with other modalities. On the other hand, combination therapies may also lead to cumulative toxicity, such as potentially lethal infections caused by the compounded risks of enhanced lymphodepletion and increased incidence of febrile neutropenia due to systemic therapies. The efficacy and safety of these combinations need to be verified through clinical trials.

Future directions and challenges

The clinical success of UCAR-T therapies in hematologic malignancies has validated the engineering principle of disrupting TCR and HLA to mitigate GvHD and HvGR. However, lessons from these trials highlight that transient efficacy and limited persistence remain major challenges hindering broader application. This durability gap,

particularly when compared to autologous products, is now the central focus of innovation.

To prevent GvHD, knocking out the TRAC locus remains the standard approach. While lymphodepletion is also critical for preventing host rejection, its associated toxicities and potential need for repetition are significant drawbacks. More extensive gene editing can engineer UCAR-T cells with inherent resistance to rejection, potentially reducing the need for intensive lymphodepletion regimens. However, as strategies to overcome complex rejection mechanisms require multiplex editing, the reliance on CRISPR-Cas systems increases the risk of chromosomal translocations resulting from simultaneous double-strand breaks. In this context, base editors offer a significant advantage by inducing targeted nucleotide changes without creating these breaks, thereby reducing genomic instability. This gentler process may also preserve cell fitness, leading to improved potency and manufacturing yields. Nevertheless, a key safety concern for base editing is the potential for sgRNA-independent, off-target mutations. Conversely, if future strategies require the insertion of large genetic payloads, CRISPR-based systems retain a key advantage due to their capacity for gene integration.

UCAR-T cells currently face challenges of limited durability and depth of remission. Research into their intrinsic properties suggests that optimizing gene editing, modifying cytokine support, and reprogramming cell phenotypes may enhance their in vivo function and persistence. In treating solid tumors, UCAR-T cells face hurdles similar to their autologous counterparts: a tumor microenvironment that hinders infiltration and promotes T cell exhaustion, and antigen heterogeneity that drives relapse. Encouragingly, because these challenges are shared, many advanced strategies developed for autologous CAR T can be directly translated and tested on UCAR-T platforms. These strategies include degrading the tumor extracellular matrix, enhancing chemokine-mediated infiltration, depleting or reprogramming immunosuppressive cell populations, and engineering UCAR-T cells to resist exhaustion.

Clinically, the reliance on intensified lymphodepletion regimens to support UCAR-T expansion creates a precarious balance between enhancing efficacy and increasing the risk of severe, prolonged cytopenias and opportunistic infections. The off-the-shelf nature of UCAR-T cells creates unique opportunities for combination with other novel modalities, such as BiTEs. This synergistic approach is already supported by promising early clinical data from trials involving autologous CAR T and CAR-NK cells. While HLA matching presents a direct strategy to mitigate both GvHD and HvGR, it substantially increases the manufacturing cost and

complexity for a truly 'off-the-shelf' product. However, should rigorous, proof-of-concept prospective studies confirm that HLA matching enhances therapeutic efficacy, the field may advance towards establishing banks of allogeneic UCAR-T cells derived from donors with diverse HLA types. Additionally, strategies such as novel delivery routes, may help penetrate the hostile immune microenvironment, yet their clinical effectiveness must be substantiated by more extensive research. Although UCAR-T cells are 'off-the-shelf' products, they differ from conventional biological products in one critical aspect: their persistence exhibits significant inter-patient variability. This heterogeneity is driven by both manufacturing techniques and HvGR. How to appropriately characterize and address this pharmacokinetic unpredictability within a regulatory framework is a issue that warrants further attention.

Conclusion

UCAR-T cell therapy addresses the manufacturing challenges of autologous products, with preclinical strategies focused on mitigating GvHD and HvGR through genetic modification of $\alpha\beta$ T cells or the use of alternative cell sources. However, key obstacles to broader clinical implementation remain, including poor in vivo persistence and suboptimal depth of remission. These challenges are compounded in solid tumors by antigen heterogeneity and the immunosuppressive tumor microenvironment. To overcome these hurdles, current research is exploring diverse strategies, many of which are informed by advances in the autologous CAR T field. Clinical strategies, including intensive detoxification, combination therapy, repeated administration, and local injection, represent promising avenues for further research and development.

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Authors' contributions

Conceptualisation, NL, SHW; investigation, NJ, ZYY, HLM, SJX, SHW, NL; writing—original draft, NJ, ZYY, HLM, SJX, SHW, NL; writing—review & editing, NJ, ZYY, HLM, SJX, SHW, NL. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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